

RANJITH KUMAR MUMMADI

Data Scientist | Machine Learning & Generative AI

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PROFESSIONAL SUMMARY

Data Scientist and Computer Science graduate with hands-on experience across machine learning, deep learning, and generative AI. Proven ability to build end-to-end data science solutions — from data cleaning and exploratory analysis to model training, evaluation, and deployment. Skilled in classical ML algorithms, computer vision (CNNs), NLP, LLMs, and RAG pipelines, with strong Python and SQL foundations. Fast learner known for translating unstructured business problems into deployable, data-driven solutions.

TECHNICAL SKILLS

Programming & Querying: Python, SQL, REST APIs

Machine Learning: Linear/Logistic Regression, Decision Tree, Random Forest, SVM, KNN, K-Means, PCA, Feature Engineering, Model Evaluation (Accuracy, Precision/Recall, F1, ROC-AUC, Cross-Validation)

Deep Learning & Computer Vision: CNN, TensorFlow, Keras, OpenCV, Image Classification, Model Tuning

NLP & Generative AI: LLMs (OpenAI GPT-3.5/4, Claude, Gemini), Prompt Engineering, RAG, LangChain, LlamaIndex, Embeddings, Vector Databases (FAISS, Pinecone, Chroma)

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, EDA, Data Cleaning & Preprocessing, Statistical Analysis

Tools & Platforms: Jupyter, Git/GitHub, Docker, FastAPI, Streamlit, Postman, Jenkins/CI-CD

PROJECT EXPERIENCE

Deep Learning Developer — Image Classification (Cat vs. Dog)

2024

Python • TensorFlow/Keras • CNN • OpenCV • NumPy • Matplotlib

- Built and trained a Convolutional Neural Network (CNN) from scratch to classify images of cats and dogs, tuning architecture and hyperparameters to improve validation accuracy.
- Designed an image preprocessing and augmentation pipeline (resizing, normalization, rotation/flip augmentation) using OpenCV and Keras ImageDataGenerator to reduce overfitting and boost generalization.
- Evaluated model performance using accuracy and loss curves and a confusion matrix; visualized results with Matplotlib to guide iterative model improvements.

Machine Learning Developer — Chronic Kidney Disease Prediction

2024

Python • Scikit-learn • Pandas • NumPy • Matplotlib

- Built an end-to-end ML classification pipeline to predict chronic kidney disease from patient clinical data, covering data cleaning, missing-value handling, feature engineering, and EDA.
- Trained and compared Logistic Regression, Decision Tree, and Random Forest models; evaluated Accuracy, Precision, Recall, and F1-score to select the best-performing model.
- Applied feature importance analysis to improve model interpretability for clinical decision support.

GenAI Application Developer — AI Resume Analyzer

2024

Python • OpenAI API • Prompt Engineering • NumPy • Pandas

- Built an AI-powered tool end-to-end — REST API integration, LLM inference, and structured JSON output — to extract skills, score candidate fit, and generate actionable feedback automatically.
- Benchmarked response quality and latency across OpenAI, Claude, and Gemini backends; logged structured experiment results to guide model selection.

EDUCATION

B.Tech — Computer Science & Engineering, Dr. M.G.R. Educational & Research Institute

Aug 2021 – May 2025

CGPA: 7.8 / 10

CERTIFICATION

Dhaapps Edutech — AI / ML Certification

SOFT SKILLS

Problem Solving | Quick Learner | Independent Worker | Stakeholder Collaboration | Analytical Thinking | Clear Documentation & Reporting